

ARTEM ZHDANOV

Senior Software Engineer

Grenoble, France | a.zhdanov@gmail.com | +33 07 43 75 43 50 | [linkedin.com/in/zhdanovartem](https://www.linkedin.com/in/zhdanovartem)

PROFESSIONAL SUMMARY

Software engineer with 5 years of production experience delivering enterprise-scale systems: ERP integrations, carbon accounting infrastructure, and cloud-connected platforms supporting national operations. Currently in Grenoble on a research exchange, completing graduate coursework in artificial intelligence and cloud computing within the MSc Informatics (MoSIG) programme at Université Grenoble Alpes. Combines industrial software delivery (Java, Python, C/C++, REST APIs, CI/CD, AWS) with a developing specialization in software development applied to energy and environmental systems. Trilingual in English, French, and Russian. Seeking a summer research internship and permanent engineering roles in the Grenoble ecosystem.

TECHNICAL SKILLS

Programming Languages: Python, Java, C, C++, C#, JavaScript, TypeScript, SQL, R, Apex, Bash

Machine Learning & Data Science: PyTorch, Scikit-learn, NumPy, Pandas, LLM application design, retrieval-augmented generation (RAG), prompt engineering, data pipelines, ETL

Scientific & High-Performance Computing: Numerical methods, linear algebra, parallel and concurrent programming, operating systems internals, algorithm optimization, Linux/Unix, image processing

Backend & Integration: REST and SOAP APIs, FastAPI, Node.js, ASP.NET Core, microservices, system integration, SAP S/4HANA interfaces, message-driven architectures

Frontend: React, Angular, TypeScript, HTML5, CSS3, Bootstrap, Lightning Web Components

Cloud & DevOps: AWS, Azure, Salesforce Platform, Git, GitHub, GitLab, CI/CD, GitOps, release management, Agile, Scrum

Databases & BI: PostgreSQL, SQL Server, MongoDB, Power BI, Tableau

Domain Expertise: Carbon accounting (GHG Scopes 1, 2, 3), ESG and sustainability reporting, energy systems data, environmental analytics, enterprise logistics

EDUCATION

Université Grenoble Alpes

Sept 2026 – June 2027

Exchange Year — Graduate Coursework in Informatics · Grenoble, France

- Exchange year taking graduate coursework within the Master of Science in Informatics (MoSIG) programme
- Coursework spanning visual computing, artificial intelligence, programming languages, and compiler design

Toronto Metropolitan University

Jan 2025 – Dec 2027

Bachelor of Science (Honours), Computer Science · Toronto, Canada

- Named to the 2025 Dean's List, Faculty of Science, in recognition of academic performance
- In-depth study of software development, data structures and algorithms, operating systems, concurrent programming, and data analysis

Niagara College

Sept 2019 – Dec 2021

Advanced Diploma, Computer Programming and Analysis (Co-op) · Welland, Canada

- Placed on the President's Honour Roll for outstanding academic achievement across five academic terms
- Recipient of the Keith Segal International Student Scholarship (2021), awarded to international students with the highest GPAs in their School of Study

PROFESSIONAL EXPERIENCE

Technology Developer — Purolator Inc.

Feb 2022 – Aug 2026

Mississauga, Canada

- Led replacement of a business-critical SQL Server legacy interface inside an ERP integration, designing an in-house solution that cut server licensing costs by 30% and reduced invoice processing errors by 75%
- Surfaced \$3M in previously missed revenue by introducing real-time error logging and automated reconciliation across the integration pipeline

- Architected Scope 1, 2, and 3 carbon accounting data models for the Net Zero Cloud sustainability programme, deploying 400+ custom components across 7 production releases and improving emissions auditability and executive visibility
- Built and deployed production AI agents with custom prompt engineering for Sales and Customer Care teams, accelerating data retrieval and task automation and raising team productivity by 25%
- Engineered full-stack, enterprise-grade solutions across the complete application lifecycle: custom user interfaces, backend services, and REST-based integrations with AWS and external platforms
- Directed development of corporate-wide systems including EDI configuration management, shipping-channel data tooling, CX survey applications, chatbot integrations, and an Enterprise Service Directory
- Owned end-to-end CI/CD and release management across onshore and offshore codebases, handling 400+ Git merge requests and 12+ production deployments annually with zero production incidents

Technologies: Salesforce, Apex, Java, Python, JavaScript, AWS, REST/SOAP APIs, SQL Server, SAP S/4HANA, Git, Postman

Computer Programmer, Research Assistant — Niagara College Research & Innovation Mar 2021 - Dec 2021

Welland, Canada

- Engineered full-stack applications for large-scale soil and mineral analysis, delivering interactive data visualizations that supported applied research and decision-making
- Optimized data pipelines and database queries, improving query performance by up to 40% and reducing storage costs by 25% across large agricultural datasets
- Accelerated delivery of a customer-facing web platform by building reusable responsive UI components and collaborating with research stakeholders on prototyping and validation, resolving 50+ defects and shipping 2 months ahead of schedule

Technologies: Python, C#, .NET, PostgreSQL, MongoDB, Angular, TypeScript, SQL, Bootstrap

SELECTED PROJECTS

Emissions Forecasting & Climate Policy Assistant

In development

- Full-stack climate analytics platform making Canada's national GHG inventory queryable and forecastable through natural language, built on ECCC and Statistics Canada open data
- Architecture: PyTorch forecasting models with classical baselines and LMDI decomposition, FastAPI backend, LangChain RAG over policy documents, PostgreSQL/pgvector storage, AWS deployment with GitHub Actions CI/CD

CERTIFICATIONS

Salesforce Certified Advanced Administrator (2023, credential 3992685)

Salesforce Certified Administrator (2022, credential 2811244)

LANGUAGES

English — native/bilingual

French — fluent, professional working proficiency

Russian — native

Spanish — limited working proficiency

ADDITIONAL

Available for a full-time research internship, March–August 2027. Convention de stage available through Université Grenoble Alpes.